

Regression Analysis Report

1. Introduction

This project aims to develop a Simple Linear Regression model to predict stock closing prices using opening prices. Linear regression is a supervised machine learning algorithm that models the relationship between an independent variable and a dependent variable. In this analysis, the Stock Prices dataset was used, and the model was evaluated using the R-squared (R^2) score and Mean Squared Error (MSE).

2. Objective

The objectives of this project are:

- To load and inspect the Stock Prices dataset.
- To split the dataset into training and testing sets.
- To build a Simple Linear Regression model using Scikit-learn.
- To predict stock closing prices based on opening prices.
- To evaluate the model using R^2 Score and Mean Squared Error (MSE).
- To interpret the regression coefficients.

3. Dataset Description

The Stock Prices dataset contains historical stock market information, including opening price, closing price, high price, low price, trading volume, and other financial indicators.

For this regression analysis:

- **Independent Variable (X):** Opening Price (open)
- **Dependent Variable (y):** Closing Price (close)

Rows containing missing values in these variables were removed before model training.

4. Data Preprocessing

Before training the model, the dataset was inspected for missing values. Missing values in the selected variables were removed using the `dropna()` function. Only the open and close columns were selected because the objective was to perform simple linear regression.

The dataset was then divided into:

- Training Set: 80%
- Testing Set: 20%

using the `train_test_split()` function.

5. Methodology

A Simple Linear Regression model from the Scikit-learn library was used for prediction.

The workflow consisted of:

1. Loading the dataset
2. Data inspection
3. Handling missing values
4. Selecting predictor and target variables
5. Splitting the dataset
6. Training the Linear Regression model
7. Making predictions
8. Evaluating the model
9. Visualizing the regression results

6. Model Evaluation

Mean Squared Error: 2.7260823685319755
Root Mean Squared Error: 1.651085209349286
R-squared Score: 0.9997383343436389

The performance of the Simple Linear Regression model was evaluated using the Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2) score. The model achieved an MSE of 2.7261 and an RMSE of 1.6511, indicating that the prediction errors are relatively

small. Additionally, the R^2 score of 0.9997 shows that approximately 99.97% of the variation in the closing stock price is explained by the opening stock price. This exceptionally high R^2 value demonstrates that the model fits the data extremely well and provides highly accurate predictions, confirming a very strong linear relationship between the opening and closing stock prices.

7. Regression Equation

Regression Equation: $\text{Closing Price} = 0.0267 + (0.9999 \times \text{Opening Price})$

The regression equation obtained from the model is:

$$\text{Closing Price} = 0.0267 + (0.9999 \times \text{Opening Price})$$

This equation indicates a strong positive linear relationship between the opening and closing stock prices. The coefficient of 0.9999 means that for every one-unit increase in the opening price, the closing price is expected to increase by approximately 0.9999 units. The intercept of 0.0267 represents the predicted closing price when the opening price is zero, although it has limited practical significance in this context. Overall, the equation shows that the opening price is an excellent predictor of the closing price.

8. Results and Visualizations

First Five Rows of the Dataset

	symbol	date	open	high	low	close	volume
0	AAL	2014-01-02	25.0700	25.8200	25.0600	25.3600	8998943
1	AAPL	2014-01-02	79.3828	79.5756	78.8601	79.0185	58791957
2	AAP	2014-01-02	110.3600	111.8800	109.2900	109.7400	542711
3	ABBV	2014-01-02	52.1200	52.3300	51.5200	51.9800	4569061
4	ABC	2014-01-02	70.1100	70.2300	69.4800	69.8900	1148391

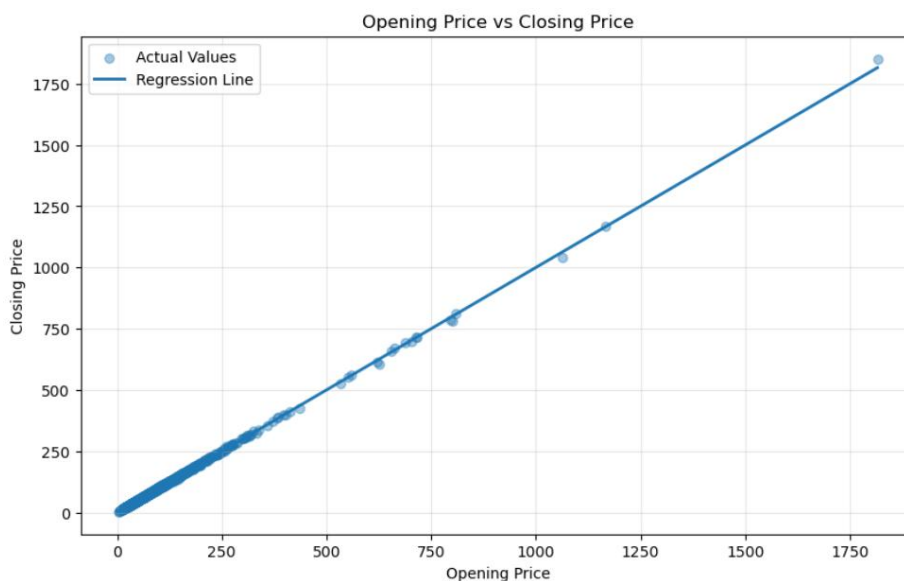
The first five rows of the dataset provide an overview of the stock price records for different companies on 2 January 2014. Each record includes the stock symbol, trading date, opening price, highest price, lowest price, closing price, and trading volume. The sample shows variations in stock prices and trading volumes across different companies, indicating that the dataset contains detailed historical stock market information suitable for analyzing the relationship between opening and closing prices using regression analysis.

Missing values summary

```
Missing values in each column:  
symbol      0  
date        0  
open       11  
high        8  
low         8  
close       0  
volume      0  
dtype: int64
```

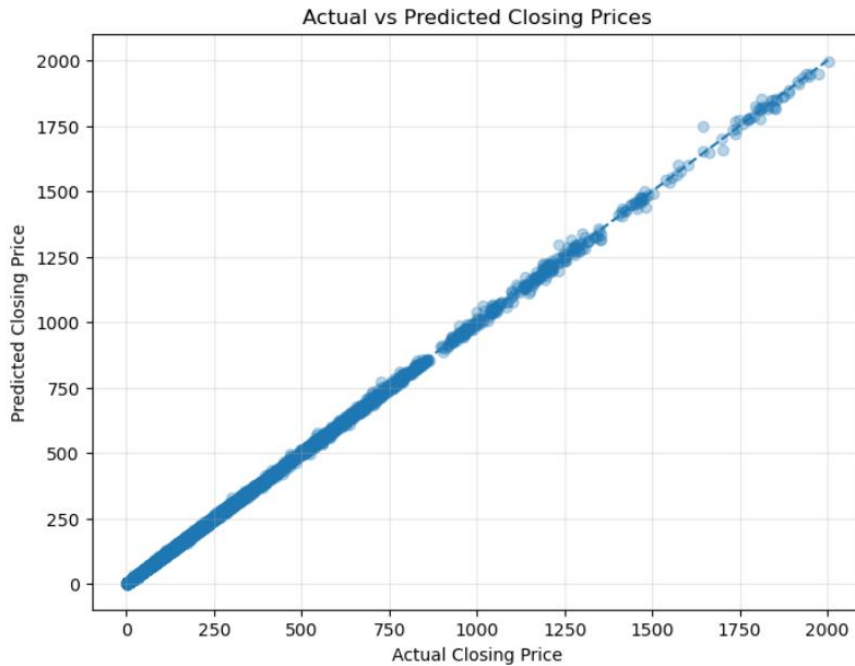
The missing values summary shows that the dataset is largely complete, with no missing values in the symbol, date, close, and volume columns. However, the open column contains 11 missing values, while the high and low columns each contain 8 missing values. Since the regression model uses the open and close variables, rows with missing values in these columns were removed during data preprocessing to ensure accurate model training and reliable prediction results.

Regression Line Plot



The regression line plot illustrates a strong positive linear relationship between the opening and closing stock prices. The data points are closely clustered around the fitted regression line, indicating that the linear regression model accurately captures the relationship between the two variables. This close alignment suggests that the opening price is an excellent predictor of the closing price, demonstrating the model's high predictive accuracy.

Actual vs Predicted Values



The Actual vs Predicted Values plot compares the model's predicted closing prices with the actual closing prices from the test dataset. Most of the data points are closely aligned along the diagonal reference line, indicating that the predicted values closely match the actual values. This demonstrates that the Simple Linear Regression model has excellent predictive accuracy and is highly effective at estimating closing stock prices based on opening prices.

Findings

- The **opening price** and **closing price** exhibit a very strong positive linear relationship, making the opening price an effective predictor of the closing price.
- Missing values in the selected variables were successfully handled by removing incomplete records, ensuring a clean dataset for model training.
- The Simple Linear Regression model achieved an **R² score of 0.9997**, indicating that approximately **99.97% of the variation in closing prices is explained by opening prices**.
- The model produced a **low Mean Squared Error (2.7261)** and **Root Mean Squared Error (1.6511)**, demonstrating high prediction accuracy and minimal prediction errors.
- The regression coefficient (**0.9999**) shows that the closing price increases almost proportionally with the opening price, confirming a nearly one-to-one relationship.
- The regression line plot and the Actual vs Predicted Values plot further confirm that the model fits the data exceptionally well, with predicted values closely matching the actual closing prices.

Conclusion

This project successfully developed a **Simple Linear Regression** model to predict stock closing prices using opening prices from the Stock Prices dataset. After preprocessing the data by removing missing values, the dataset was divided into training and testing sets, and the model was trained using Scikit-learn. The evaluation results, including an **R² score of 0.9997** and a **low Mean Squared Error of 2.7261**, indicate that the model provides highly accurate predictions. The regression equation and visualizations further confirmed a strong positive linear relationship between the opening and closing prices. Overall, the results demonstrate that Simple Linear Regression is an effective and reliable method for predicting stock closing prices based on opening prices.